



Application Note

MI Data - P4-2v HIQ Scripts

Peng Jiang
Verasonics Inc.

November 2019

Table of Contents

Document Change History	2
1. Disclaimer	2
2. Measurement Information	3
3. MI Data	3
3.1 P4-2v 128RyLnsPolar	3
3.2 P4-2v WideBeamPolar	4
3.3 P4-2v WideBeamHIPolar (Inversed transmit).....	6
3.4 P4-2v WideBeamDoppler (D-mode).....	7

Document Change History

Date	Section(s)	Description	Author
November 6, 2019	All	Rev A: Initial document	PJ

1. Disclaimer

The MI data provided below are for reference only. It is the user's responsibility to determine whether to perform additional measurements depending on his or her use of the information.

The measurements were taken using one transducer on one Vantage System. Verasonics did not evaluate variability between different transducers nor between different Vantage Systems, so the MI values may be different if the measurements were conducted with a different transducer and/or Vantage System. Moreover, measurement errors may be introduced when using a different hydrophone or measurement system.

The measurement system and measurement procedure are developed by Verasonics following NEMA and IEC standards.

1. AIUM NEMA UD2 Acoustic output measurement standard for diagnostic ultrasound equipment. 2007.
2. IEC 62359 Ultrasonics - Field characterization - Test methods for the determination of thermal and mechanical indices related to medical diagnostic ultrasonic fields. Ed. 2.0. 2010.

3. AIUM NEMA UD3 Standard for real time display of thermal and mechanical acoustic output indices on diagnostic ultrasound equipment. 2004 – Revision 2.

2. Measurement Information

P4-2v HIQ script Mechanical Index (MI) values were measured with

Transducer: S/N NDN79020815
 System: S/N KV22446
 Hydrophone: S/N D1604-117 (Precision Acoustics, Calibrated)

Note:

1. TxVoltage in the data listed below is the transmit voltage entered in the GUI.
2. Location is the position in acoustic field where the MI data was measured, where Z is the depth from transducer surface, XZ is the imaging plane and YZ is the elevation plane. (0,0,0) is the location of geometry center of a transducer.

3. MI Data

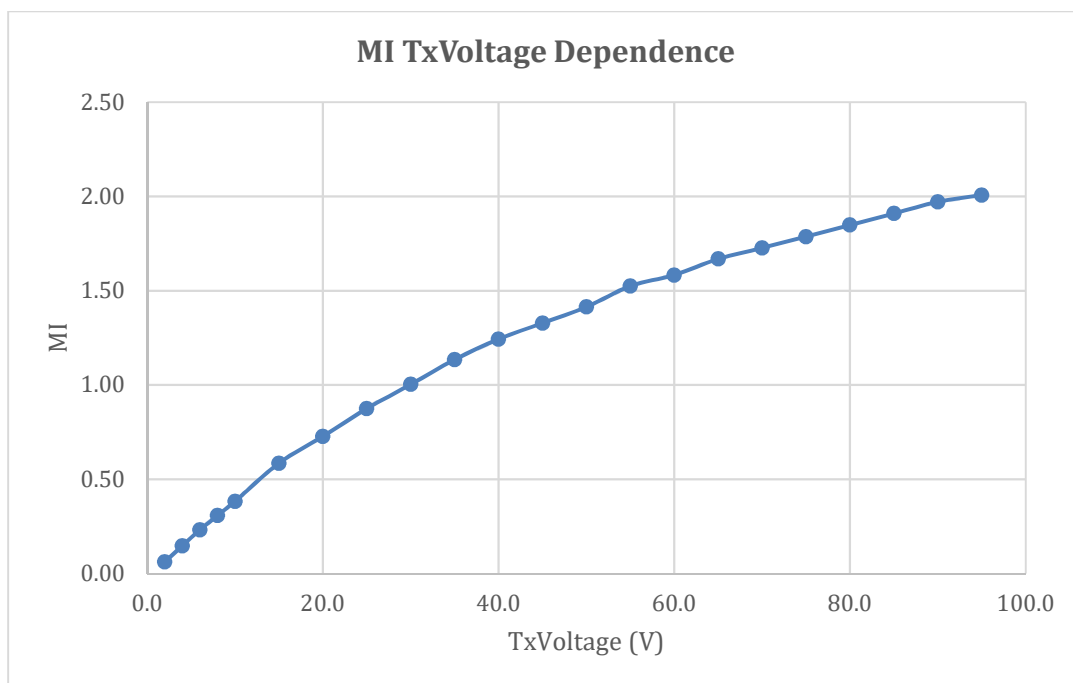
3.1 P4-2v 128RyLnsPolar

Software release: 4.1.100-1812101200

Location: (0.00, 0.00, 49.87) mm

TxVoltage (V)	Pp (Mpa)	Pr (Mpa)	MI
2.0	0.21	0.17	0.06
4.0	0.52	0.41	0.15
6.0	0.85	0.65	0.23
8.0	1.19	0.86	0.31
10.0	1.57	1.06	0.38
15.0	2.57	1.63	0.59
20.0	3.66	2.01	0.73
25.0	4.82	2.40	0.88
30.0	5.60	2.72	1.00
35.0	5.79	3.04	1.13
40.0	5.92	3.30	1.24
45.0	5.98	3.50	1.33

50.0	5.98	3.69	1.41
55.0	5.98	3.95	1.53
60.0	5.98	4.08	1.58
65.0	5.98	4.27	1.67
70.0	6.03	4.39	1.73
75.0	5.97	4.52	1.79
80.0	5.97	4.64	1.85
85.0	5.97	4.77	1.91
90.0	5.97	4.90	1.97
95.0	5.97	4.97	2.01



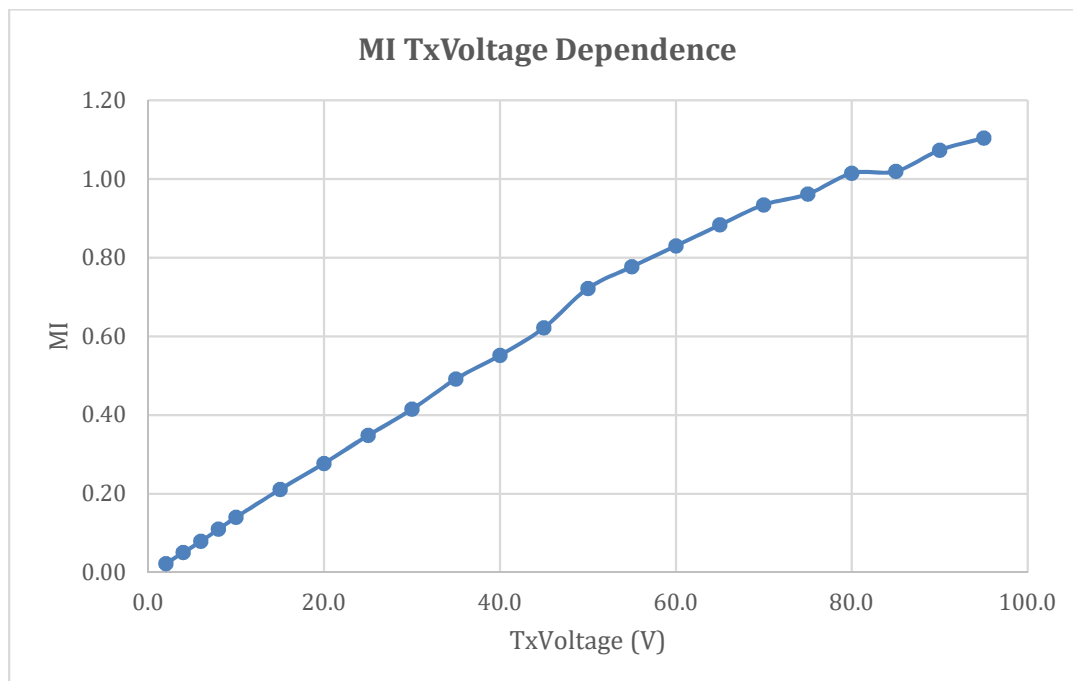
3.2 P4-2v WideBeamPolar

Software release: 4.1.100-1812101200

Location: (-4.00, 0.00, 50.03) mm

TxVoltage (V)	Pp (Mpa)	Pr (Mpa)	MI
2.0	0.06	0.06	0.02
4.0	0.15	0.14	0.05
6.0	0.23	0.22	0.08

8.0	0.33	0.30	0.11
10.0	0.42	0.39	0.14
15.0	0.66	0.58	0.21
20.0	0.90	0.76	0.28
25.0	1.15	0.96	0.35
30.0	1.39	1.14	0.41
35.0	1.65	1.34	0.49
40.0	1.88	1.50	0.55
45.0	2.29	1.68	0.62
50.0	2.76	1.94	0.72
55.0	3.15	2.07	0.78
60.0	3.54	2.20	0.83
65.0	3.86	2.33	0.88
70.0	4.12	2.46	0.93
75.0	4.38	2.52	0.96
80.0	4.63	2.65	1.01
85.0	4.76	2.65	1.02
90.0	4.96	2.78	1.07
95.0	5.09	2.85	1.10

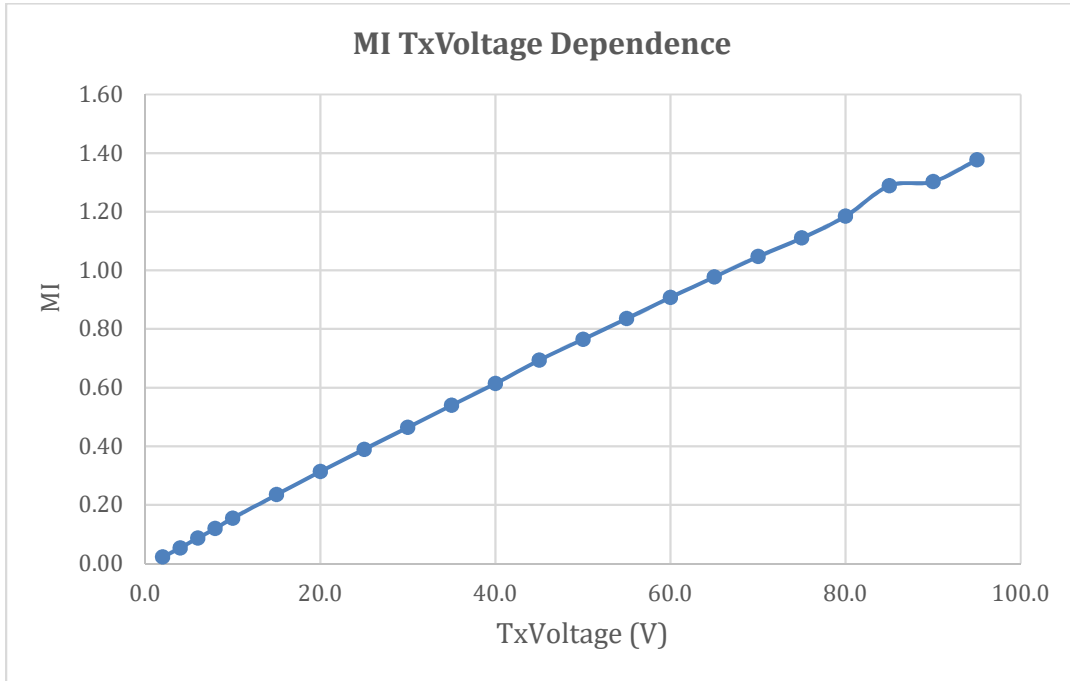


3.3 P4-2v WideBeamHIPolar (Inversed transmit)

Software release: 4.1.100-1812101200

Location: (0.00, 0.00, 42.15) mm

TxVoltage (V)	Pp (Mpa)	Pr (Mpa)	MI
2.0	0.04	0.04	0.02
4.0	0.10	0.11	0.05
6.0	0.16	0.18	0.09
8.0	0.22	0.24	0.12
10.0	0.27	0.31	0.15
15.0	0.43	0.47	0.23
20.0	0.57	0.63	0.31
25.0	0.72	0.79	0.39
30.0	0.88	0.94	0.46
35.0	1.03	1.09	0.54
40.0	1.19	1.24	0.61
45.0	1.34	1.39	0.69
50.0	1.50	1.53	0.76
55.0	1.66	1.67	0.84
60.0	1.83	1.81	0.91
65.0	1.97	1.94	0.98
70.0	2.13	2.08	1.05
75.0	2.28	2.20	1.11
80.0	2.47	2.34	1.18
85.0	2.64	2.54	1.29
90.0	2.79	2.67	1.30
95.0	2.98	2.81	1.38



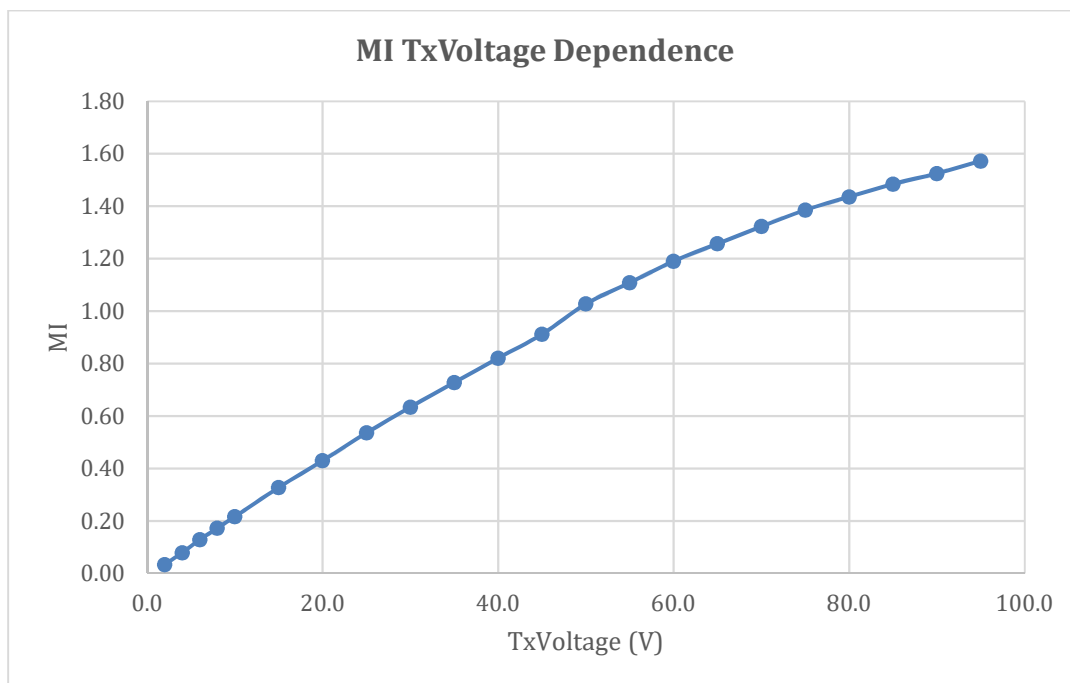
3.4 P4-2v WideBeamDoppler (D-mode)

Software release: 4.1.100-1812101200

Location: (6.00, 0.00, 38.83) mm

TxVoltage (V)	Pp (Mpa)	Pr (Mpa)	MI
2.0	0.08	0.08	0.03
4.0	0.18	0.18	0.08
6.0	0.29	0.29	0.13
8.0	0.40	0.39	0.17
10.0	0.51	0.49	0.22
15.0	0.78	0.74	0.33
20.0	1.05	0.97	0.43
25.0	1.33	1.21	0.54
30.0	1.61	1.43	0.63
35.0	1.89	1.64	0.73
40.0	2.15	1.85	0.82
45.0	2.41	2.05	0.91
50.0	2.68	2.31	1.03
55.0	3.23	2.49	1.11

60.0	3.80	2.68	1.19
65.0	4.35	2.83	1.26
70.0	4.84	2.97	1.32
75.0	5.25	3.12	1.39
80.0	5.46	3.23	1.44
85.0	5.60	3.34	1.48
90.0	5.66	3.43	1.53
95.0	5.70	3.53	1.57



This document is intended for use by Verasonics customers only.

Do not duplicate or distribute without permission.

Verasonics, Inc.
11335 NE 122nd Way, Suite 100
Kirkland, WA 98034
USA